

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
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MINI-PROJECT (BEC586) REPORT

on

“ VOICE CONTROLLED CAR AND OBSTACLE DETECTION ”

Submitted in partial fulfillment of the requirements for the award of the degree

BACHELOR OF ENGINEERING
IN
ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

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CERTIFICATE

Certified that the Mini-project (BEC586) work entitled, “VOICE CONTROLLED CAR AND OBSTACLE DETECTION ”carried out by Mr. SHRIDHAR SHIVARAY KURIHULI (1MV23EC106), SRUJAN G (1MV23EC109), SUHAS G (1MV23EC112),YASHWANTH H S (1MV23EC125) the bonafide students of Sir M. Visvesvaraya Institute of Technology in partial fulfillment for the award of Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belgaum during the year 2025-26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements for Mini-Project work prescribed for the said Degree.

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DECLARATION

We, students of the 5th semester hereby declare that the project report on, “**VOICE CONTROLLED CAR AND OBSTACLE DETECTION**” has been presented under the guidance of Mrs. Praveena N , Assistant Professor, Department of Electronics and Communication Engineering, Sir M. Visvesvaraya Institute of Technology, Bengaluru as partial fulfillment of the requirement for the award of Bachelor of Engineering in Electronics & Communication Engineering by Visvesvaraya Technological University, Belagavi during the academic year 2025-2026. This topic has not been submitted previously for any degree or diploma of any institution.

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ABSTRACT

The integration of intelligent control systems in modern robotics has significantly enhanced the way autonomous vehicles operate. This mini project, **“Voice Controlled Car and Obstacle Detection,”** focuses on designing and developing a compact robotic vehicle capable of responding to human voice commands while simultaneously ensuring safe navigation through real-time obstacle detection. The primary objective of this project is to create a user-friendly and interactive system that simplifies vehicle control, especially in applications where manual operation is difficult or inefficient.

The system utilizes a voice recognition module to interpret user commands such as “forward,” “backward,” “left,” “right,” and “stop.” These commands are processed by a microcontroller, which then actuates the motor driver to control the movement of the car. In parallel, the vehicle is equipped with an ultrasonic sensor that continuously monitors the surroundings to detect obstacles within a predefined range. When an obstacle is detected, the system either alerts the user or automatically stops or redirects the vehicle to prevent collisions. This dual-function design improves both the usability and safety of the vehicle.

The project demonstrates effective integration of speech recognition, sensor technology, and embedded control to achieve semi-autonomous navigation. It can be applied in various fields, including assistive robotics, smart mobility solutions, and automation systems. The results showed that the vehicle responded accurately to voice commands and successfully detected obstacles in its path, ensuring safe operation. Overall, the project highlights the potential of combining human-machine interaction with intelligent sensing to enhance robotic mobility and user convenience.

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Chapter 1: INTRODUCTION

In recent years, the rapid advancement of automation, embedded systems, and human-machine interaction has significantly transformed the way we control and operate machines. One of the emerging trends in this field is the integration of **voice recognition technology** with autonomous systems to create smarter, more intuitive, and user-friendly devices. The project titled “**Voice Controlled Car and Obstacle Detection**” is an innovative attempt to combine **speech-based control** with **real-time environmental sensing** to develop a compact, intelligent robotic vehicle capable of responding to user commands while ensuring safe

Voice control provides a natural and hands-free method of communication, allowing users to operate the car with simple verbal instructions such as “forward,” “left,” or “stop.” This enhances accessibility, especially for individuals who face difficulty with manual controls, and also opens up new possibilities for gesture-free automation. The system leverages a speech-processing module or mobile-based voice interface to interpret spoken commands, which are then processed by a microcontroller to drive the motors accordingly.

Along with voice operation, the car integrates an **obstacle detection module** using sensors such as ultrasonic or infrared detectors. This component continuously monitors the surroundings and detects any object in the vehicle’s path. Upon identifying an obstacle within a predefined threshold distance, the system automatically halts or avoids the object to prevent collisions. This ensures safe operation, improves reliability, and enhances the autonomous capability of the model.

The combination of **voice recognition** and **obstacle detection** makes the system both interactive and intelligent. It demonstrates the practical use of embedded systems, sensor integration, communication technologies, and automation principles. This project not only serves as an excellent learning platform for understanding microcontroller-based design, but also represents a simplified version of modern smart vehicles used in robotics, assistive technology, and autonomous navigation systems.

Overall, the **Voice Controlled Car and Obstacle Detection** project showcases how human-machine interaction can be made seamless and efficient by merging voice commands with real-time environmental awareness. It highlights the potential for developing safer, smarter, and more user-friendly robotic applications across various domains including personal assistance, industrial automation, and educational robotics.

Chapter 2: PROBLEM STATEMENT

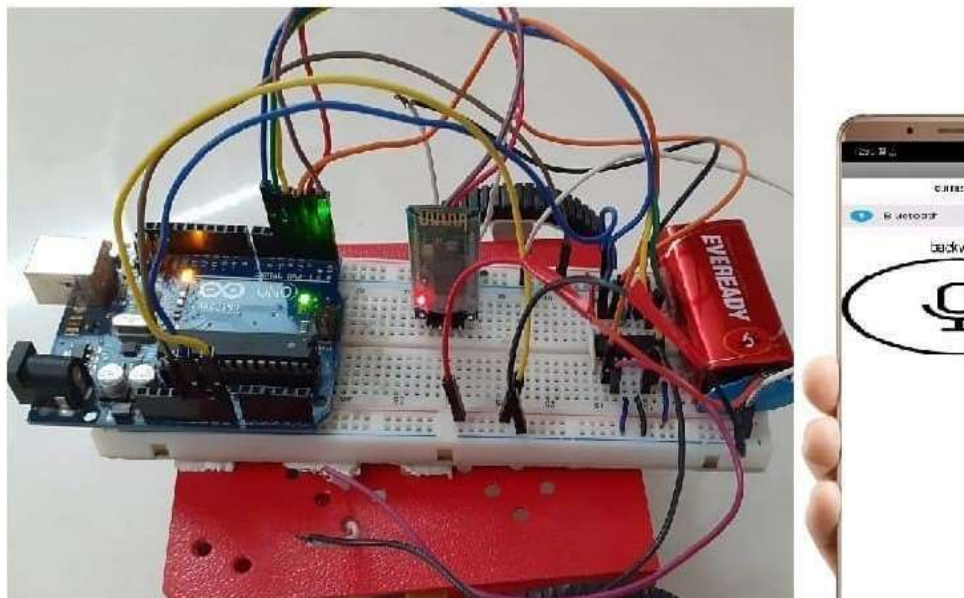
“There is a need to design a wireless voice controlled car that can be safely operated without physical interaction , allowing users to control its movement from a distance, and also avoid the obstacle.”

Chapter 3 : **LITERATURE SURVEY PAPERS**

3.1 - Survey Paper – 1 : Voice Controlled Robot Car Using Arduino

(e-ISSN:2395-0056 , Shubh Srivastava and Rajanish Singh ,May 2020)

The project focuses on creating a Voice Control Robot Car using Arduino, motor drivers, and a Bluetooth module. The hardware is designed first, followed by programming and simulation on software before interfacing with the hardware. The car is controlled via voice commands received through a Bluetooth connection with a smartphone. The project was chosen due to the growing importance of robotics in daily life and its significant role in technological advancements.



3.2 - Survey Paper – 2 : Implementation of human voice controlled robotic car

(DOI : 10.1109/SMART52563.2021.9676319 ,

RubinaLiyakat Khan , DeepaPriyanshu , Dec 2021)

This study presents a simple voice-controlled robotic car system that uses an Android app to send voice commands to an Arduino-based microcontroller via Bluetooth. The system converts spoken commands into text, which the microcontroller processes to control the car's movements, such as moving forward or turning. The project uses an ATmega Arduino board programmed in Embedded C/C++ and demonstrates an affordable, effective solution for basic robot control using voice commands.

Advantages :

- Easy hands-free control
- Safe in hazardous areas

Disadvantage :

- Limited
range
- Noise/
Voice
accuracy
issues



3.3 - Survey Paper –3 : Smartphone Controlled Multipurpose Robot Car

(Balendu Teterbay , Ayush Srivastava , Akshay Bhati , Dr.Abhay A.Deshpande

et al.

“Smart Phone Controlled Multipurpose Robot Car”. International Journal of Research & Technology (IJERT). ISSN:2278-0181 , Volume 9 Issue 05 , May-2020.)

This project focuses on creating a versatile robot car controlled via a smartphone application.

The system incorporates various sensors, including soil pH, temperature, gas, and flame sensors, to detect environmental hazards. It also features a robotic arm for pick-and-place operations and provides live video surveillance through a camera mounted on the vehicle.

The robot's Arduino, as it controls his gesture according to the orders he receives. Pause the robots "go forward", "go back", "go left", "go right" and "stop" and stop back and forth and left and right according to the voice command.



3.4 - Survey Paper –4 : Voice controlled robotic car

(issn : 2347-7180 , sandip kumar nayak,

saradaprassanahota , may

2022)

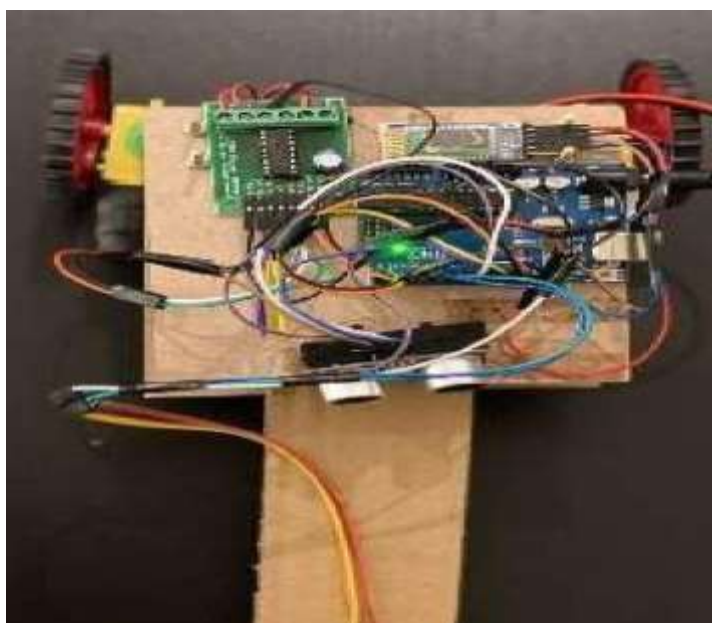
A voice-controlled robotic system uses an Android device to send voice commands via Bluetooth to an Arduino, which controls the vehicle's motors for movement in four directions. It includes a motor driver, wireless camera with night vision, robotic arm, LCD display, and ultrasonic sensor for obstacle detection, making it useful in high-risk areas for safe remote operation.

Advantages :

1. Easy, hands free voice control
2. Obstacle detection and camera

Disadvantages:

1. Low battery backup
2. load capacity

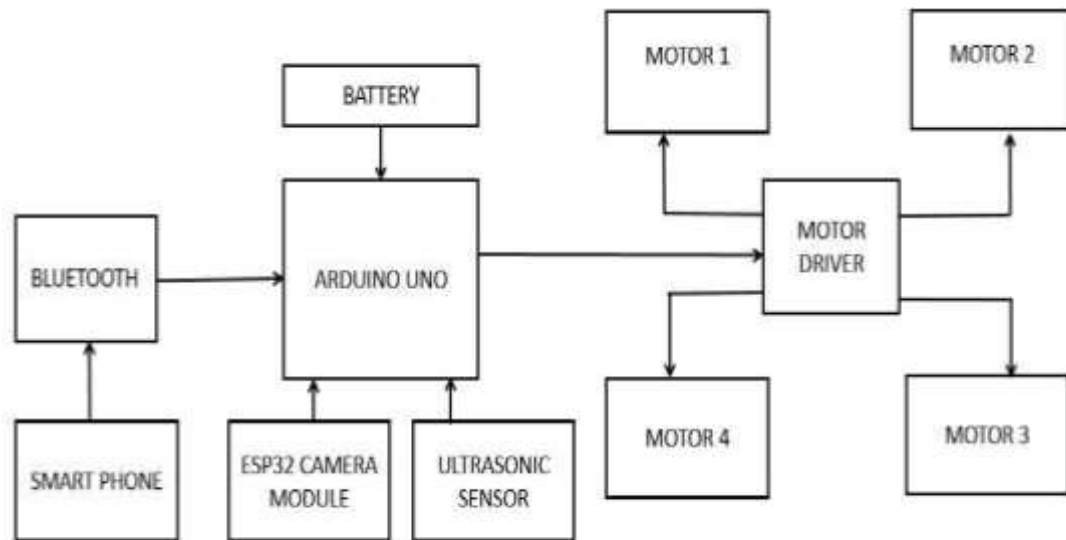


Chapter 4 :**Objectives**

- ❖ To design a car prototype controlled by voice commands.
- ❖ To establish wireless communication between the user and car , obstacle detection.
- ❖ To ensure human safety.

Chapter 5 :

Methodology



- The block diagram illustrates the working architecture of the **Voice-Controlled Car with Obstacle Detection** system.
- A **smartphone** sends voice commands through a **Bluetooth module** to control the vehicle's movement.
- The **Bluetooth module** receives these commands and forwards them to the **Arduino Uno** for processing.
- The **Arduino Uno** acts as the central controller, interpreting the received instructions.
- A **battery** powers the Arduino Uno and other connected components in the system.
- An **ultrasonic sensor** is interfaced with the Arduino to detect obstacles ahead of the vehicle.
- Based on sensor readings, the Arduino can automatically stop or change the direction to avoid collisions.
- The processed commands are sent from the Arduino Uno to the **motor driver module**.
- The **motor driver** controls four DC motors—Motor 1, Motor 2, Motor 3, and Motor 4—responsible for directional movement.
- This coordinated operation enables the car to move forward, backward, left, or right based on voice commands while ensuring obstacle avoidance.

Chapter 6 : **Hardware Details**

1. ESP32-CAM Module

- A low-cost development board with built-in camera and Wi-Fi/Bluetooth capabilities.
- Ideal for image processing, surveillance, IoT, and smart devices.

2. Bluetooth Module (HC-05 / HC-06)

- Connects smartphone (voice app) to Arduino wirelessly.
- Transfers decoded voice commands.

3. Motor Driver Module (L293D / L298N)

- Interface between Arduino and DC motors.
- Controls forward, reverse, left, right movement.

4. Ultrasonic Sensor (HC-SR04)

- A sensor that measures distance using sound waves.
- Emits ultrasonic waves and calculates the time taken for the echo to return.

5. DC Motors (4 motors with wheels)

- Provide motion to the robot car.
- Controlled via motor driver.

6. Chassis (Car Base Frame)

- Base platform for mounting components.
- Holds motors, wheels, and electronics.

7. Power Supply (Battery Pack – 9V/12V or Li-ion cells)

- Provides power to Arduino, motor driver, and motors.
- Separate regulated supply may be needed for stability.

8. Arduino Uno

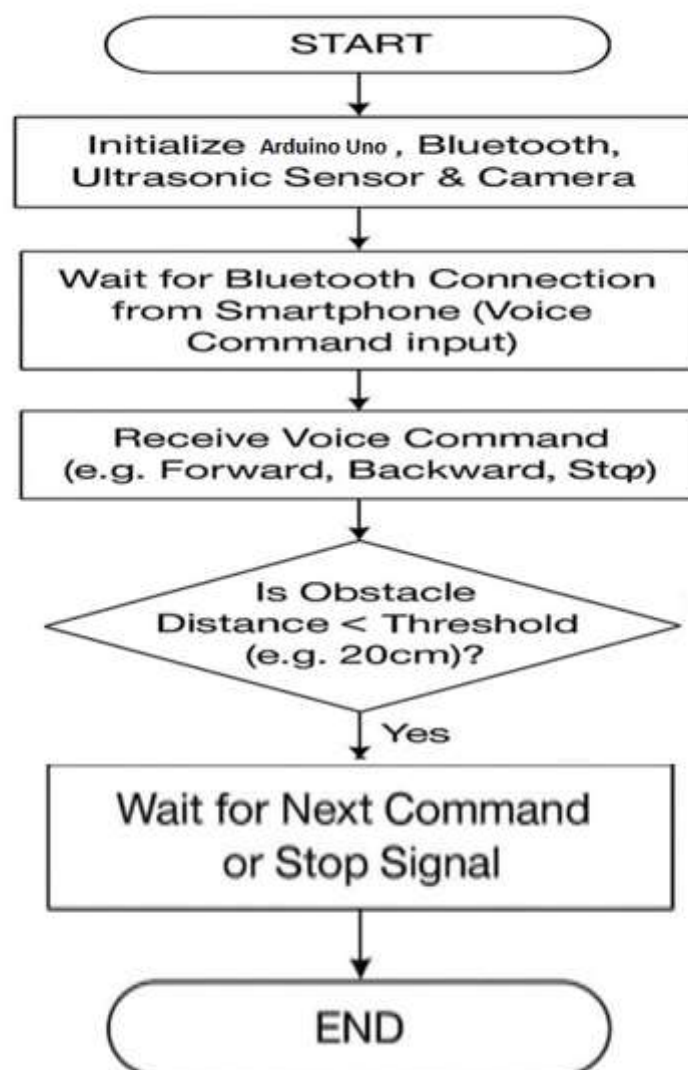
- The Arduino Uno is an open-source microcontroller board on the ATmega328P Processor.
 - Acts as an CPU for system.
-

Chapter 6 :

Software Details

- **Arduino IDE:** For programming microcontroller
- **Programming Language:** Embedded C / Arduino C++
- **Android Device:** With voice command app (Arduino Bluetooth Control)
- **Bluetooth Module (HC-05):** Wireless link

FLOW CHART



Chapter 7:

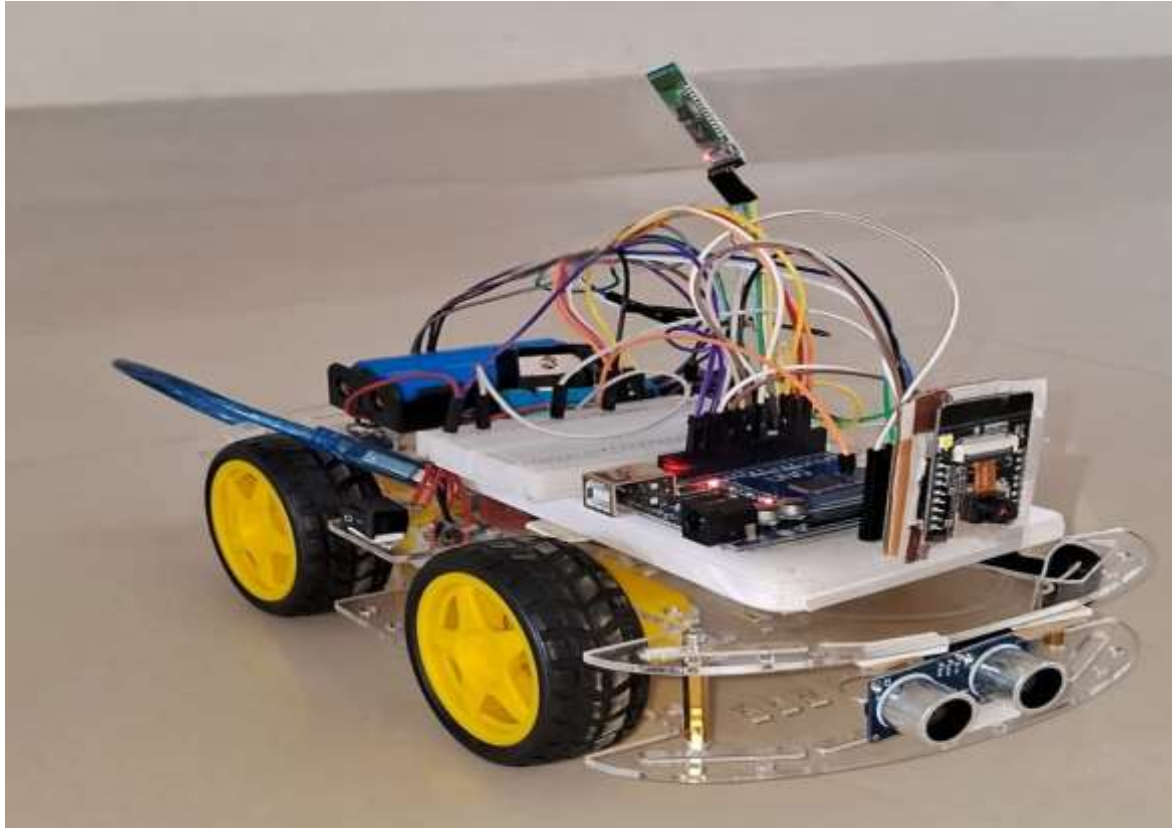
Results and Discussions

The developed **Voice-Controlled Car with Obstacle Detection** successfully responds to user voice commands for forward, backward, left, right, and stop movements with high accuracy.

The integrated **ultrasonic sensor** reliably detects obstacles within the predefined range and automatically stops the car, preventing collisions.

The integration of the **ESP32-CAM module** enables **real-time video streaming**, allowing the user to **visually monitor the car's surroundings** remotely. This improves navigation accuracy, enhances safety, and provides a live first-person view (FPV) while controlling the car through voice commands.

The system demonstrates **smooth mobility, accurate voice recognition, real-time obstacle detection, and safe autonomous control**, proving its effectiveness for smart robotic applications.



Chapter : 8 Advantages :

- **Hands-free control** using voice commands, making operation simple and user-friendly.
- **Safety enhanced** through real-time obstacle detection and automatic stopping.
- **Suitable for elderly and disabled users** who cannot operate traditional remote controls.
- **Low-cost implementation** using common components like Arduino / NodeMCU and ultrasonic sensors.
- **Wireless operation** allows control from a distance.
- **Good application in robotics, automation, and smart mobility projects.**
- **Energy efficient** due to low-power microcontroller and sensor usage.

8.1 : Limitations :

- **Voice recognition accuracy** can reduce in noisy environments.
- Requires a **stable internet connection** if using online voice-control platforms (e.g., Google Assistant).
- **Limited range of control** depending on Bluetooth/Wi-Fi module.
- **Delay in response** may occur depending on voice-processing method.
- **Not suitable for high-speed movement** due to sensor reaction time.
- **Cannot detect very small or transparent objects** reliably.
- **Battery life limitations** during extended usage.
- System performance depends on **proper calibration of sensors and motors.**

Chapter 9 :

Conclusion

- The project successfully demonstrates a **voice-operated robotic car** capable of responding accurately to user commands.
- Integrated **ultrasonic obstacle detection** ensures safe navigation by stopping the car when an obstacle is detected.
- The system proves to be **efficient, low-cost, and user-friendly**, showing strong potential for smart mobility applications.
- Overall, the project achieves its objective of combining **voice recognition technology with real-time obstacle avoidance** to create an intelligent and interactive robotic system.

9.1 : Future Scope

- Integration of **AI-based offline voice recognition** for higher accuracy even in noisy environments.
- Adding **multiple ultrasonic / IR sensors** for 360° obstacle detection.
- Implementing **GPS tracking and IoT connectivity** for remote monitoring and control through a mobile app.
- Adding **autonomous navigation** using machine learning and camera-based vision.
- Increasing range by using **Wi-Fi, LoRa, or GSM modules** instead of Bluetooth.
- Enhancing mobility with **improved motor drivers** and advanced path-planning algorithms.
- Useful in applications such as **smart wheelchairs, warehouse automation, surveillance robots**, and assistive devices.

Chapter 10 : SDGs with Justification

SDG with Icon	Justification
 The icon for SDG 3 is a green square. It features the number '3' in large white font, followed by the text 'GOOD HEALTH AND WELL-BEING' in white capital letters. Below the text is a white line graph with a heart shape at the end, symbolizing health and well-being.	<u>SDG 3 – Good Health and Well-Being</u> Justification: • Provides hands-free control, useful for people with mobility impairments, elderly users, and those unable to operate manual controls. • Reduces chances of accidents by integrating automatic obstacle detection, improving user safety. • Can be extended to assistive technologies like smart wheelchairs, medical trolleys, or support robots.
 The icon for SDG 4 is a red square. It features the number '4' in large white font, followed by the text 'QUALITY EDUCATION' in white capital letters. Below the text is a white icon of an open book and a pencil, symbolizing education.	<u>SDG 4 – Quality Education</u> Justification: • Provides students hands-on experience in robotics, embedded systems, IoT, and automation, enhancing technical learning. • Encourages project-based learning, critical thinking, and practical implementation of engineering concepts. • Serves as a model for STEM learning, helping students understand real-world applications.

References

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